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Research Paper

A comparison of harvest index estimation methods of winter wheat based on field measurements of biophysical and spectral data

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Crop harvest index (HI) is a key parameter in grain yield simulation from indirect biomass estimates and varies with crop growth conditions. How to access the HI still remains an issue to be addressed. Several HI estimation techniques for wheat have been proposed based on the fraction of water transpired after anthesis (θ_E or θ_{Ee}), the fraction of biomass accumulation after anthesis (f_G) and the ratio of the mean normalized difference vegetation index (NDVI) after anthesis to that of the pre-anthesis period ($NDVI_{post}/NDVI_{pre}$). In this paper, efforts were made to test the performances of the above HI estimation methods based on field measurements of winter wheat at different nitrogen (N) levels. Results showed that the HI changed significantly from low-N to high-N treatments, while the θ_E or θ_{Ee} was not sensitive to the N deficiency. In addition, no consistent effect of N fertilizer on f_G was observed. Using the canopy NDVI to estimate HI seemed workable since strong correlations were found between HI, the term ' $NDVI_{post}/NDVI_{pre}$ ' and the fraction of cumulative NDVI after anthesis. Further analysis suggested a good estimation of HI can be obtained just using the NDVI dataset from anthesis to maturity, especially via binary regression relations of HI, NDVI at anthesis and cumulative NDVI after anthesis. The proposed method in this study may be well oriented to practical application as its short-term dataset is relatively easy to access from satellite platforms.

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1. Introduction

Harvest index (HI), the ratio of grain yield to the total above-ground biomass (Donald & Hamblin, 1976), has been shown to be closely related to crop growing conditions (Aggarwal, Singh, Chaturvedi, & Sinha, 1986; Moser, Feil, Jampatong, & Stamp, 2006; Pirmoradian & Sepaskhah, 2006; Sinclair, 1998). Now HI has been widely used in crop yield simulation by the crop models, especially in remote estimates of crop yield from

indirect biomass estimates (Lobell, Asner, Ortiz-Monasterio, & Benning, 2003; Moriondo, Maselli, & Bindi, 2007). To obtain an accurate estimation of crop yield, the variations of HI under different environmental conditions and crop management strategies should be taken into account (Moriondo et al., 2007; Sadras & Connor, 1991). However, due to the difficulties in HI assessment, many investigations use a constant value of HI in crop yield simulation (Bastiaanssen & Ali, 2003; Duchemin, Maisongrande, Boulet, & Benhadj, 2008; Tao, Yokozawa,

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Zhang, Xu, & Hayashi, 2005). Hence, for maximizing the effective application of HI in crop yield simulation, proper HI estimation procedures are necessary (Kemanian, Stockle, Huggins, & Viegas, 2007; Sadras & Connor, 1991).

Several HI estimation approaches have been proposed. When studying leaf area development and its effect on water use, yield and the HI of water-stressed wheat, Richards and Townley-Smith (1987) found a nonlinear relationship between HI and the proportion of water transpired after anthesis:

$$HI = a + b \cdot \ln(\theta_E) \quad (1)$$

where θ_E is the ratio of the cumulative transpiration in the anthesis-maturity period to that in the emergence-maturity period, and a and b are parameters with fitted values of -0.10 and 0.13 , respectively. This relationship was obtained by regression analysis from field data of wheat under drought conditions, and hence has an empirical basis.

Based on Richards and Townley-Smith (1987), Sadras and Connor (1991) examined the empirical and theoretical aspects of the relationship between the HI and the θ_E , and developed a method to estimate HI:

$$HI_{pv} = \theta_{Ee} / [1 - (a - b \cdot \theta_{Ee})] \quad (2)$$

where subscripts pv and e indicate that the HI and θ_E are normalized by the production value and the vapour pressure deficit, respectively. Production values, i.e. the amount of biomass produced per unit of hexose substrate, may differ between the pre and post-anthesis periods for specific species. For wheat, production values do not vary substantially with ontogeny, so the normalization of HI by production value is not needed. The parameters a and b are fitted from observation data and for wheat they were determined to be 0.58 and 1.68 respectively by Sadras and Connor (1991). In addition, according to Sadras and Connor (1991), the parameters a and b have physiological meanings: a quantifies the potential contribution of pre-anthesis assimilates to yield and $1/(1-a+b)$ represents the potential HI. This method was developed in relation to water stress, and its applicability to other conditions needs to be tested.

Based on the concept described by Sadras and Connor (1991), Kemanian et al. (2007) proposed a method to estimate HI using the fraction of biomass accumulation after anthesis (f_G): f_G is defined as the ratio of aboveground biomass produced after anthesis to that for the entire growing season. Two different forms of equations are given:

$$HI = HI_0 + s \cdot f_G \quad (3)$$

$$HI = HI_X - (HI_X - HI_0) \cdot \exp(-k \cdot f_G) \quad (4)$$

where the parameters HI_0 and s are respectively the intercept and slope in the linear regression equation. In the curvilinear relationship, HI_X is the possible maximum value of HI, HI_0 is the intercept and k is a constant. A major advantage of this method is the ease of parameterization just using yield data and biomass measurements at anthesis and harvest. For wheat, the fitted values of HI_0 and s for Kemanian et al. (2007) were 0.34 and 0.21 in the linear model and HI_X , HI_0 and k in the curvilinear model were 0.52 , 0.31 and 1.99 respectively.

Spectral information from the crop canopy has been increasingly used as a tool for crop assessment (Hansen &

Schjoerring, 2003; Li et al., 2008; Rudorff & Batista, 1990; Scotford & Miller, 2004; Xue, Cao, & Yang, 2007). The most widely used vegetation index is the normalized difference vegetation index, NDVI:

$$NDVI = \frac{R_{Nir} - R_{Red}}{R_{Nir} + R_{Red}} \quad (5)$$

where R_{Nir} and R_{Red} is respectively the reflectance in the near infrared and the red band. Moriondo et al. (2007) utilized the NDVI data to estimate HI of wheat in regional yield simulation. The approach assumes that the environmental effects on the optimal harvest index (HI_{max}) are related to the reduction of NDVI from pre-anthesis to post-anthesis period. Therefore, by modifying the HI_{max} using the NDVI data collected for the entire wheat season, the actual harvest index can be obtained from Eq. (6):

$$HI = HI_{max} - HI_{range} \cdot \left(1 - \frac{NDVI_{post}}{NDVI_{pre}}\right) \quad (6)$$

where HI_{range} indicates the possible range of HI, $NDVI_{pre}$ represents the average value of NDVI from emergence to anthesis, and $NDVI_{post}$ is the mean NDVI value from anthesis to maturity. In Moriondo et al. (2007), this method was directly applied to HI estimation using the NDVI dataset from satellite platform, with HI_{max} and HI_{range} of 0.48 and 0.18 , respectively. As yet, no further investigations have tested this relationship using field data.

Moreover, the fields of application of the above proposed methods are basically different. The approaches based on transpiration and aboveground biomass may be limited to a local scale when a dense meteorological network is not available. The NDVI-based approach is more oriented to be applied on a regional scale. The objectives of this paper are to evaluate the performances of the above-mentioned HI estimation methods for winter wheat based on field measurements of biophysical and spectral data, and to explore an HI estimation approach with good feasibility and accuracy. Considering that nitrogen (N) is a major limiting factor for crop production in most agricultural areas of the world (Baethgen, Christianson, & Lamothe, 1995; Latiri-Souki, Nortcliff, & Lawlor, 1998; Li et al., 2008; Olesen, Berntsen, Hansen, Petersen, & Petersen, 2002), it was chosen for this study.

2. Materials and methods

2.1. Field experiments

The field experiments were conducted at the Yucheng Comprehensive Experimental Station (116.6°E , 36.8°N) from 2005 to 2010. The experimental station is located in the North China Plain, one of the major food-producing regions of China, with a sub-humid continental monsoon climate. The soil is mainly silt loam, with organic matter content between 0.6% and 1.0% . The growth season of winter wheat is commonly from mid-October to mid-June of the next year. The datasets used in this study were collected from two wheat seasons of 2008/2009 and 2009/2010.

The experiment consists of 15 plots of $5 \text{ m} \times 10 \text{ m}$. The plots were isolated from each other by a concrete-wall 0.1 m

wide, 1.0 m deep below the ground and 0.1 m high above the ground. Five N fertilization levels were respectively nil (N0), 70 kg [N] ha⁻¹ (N70), 140 kg [N] ha⁻¹ (N140), 210 kg [N] ha⁻¹ (N210) and 280 kg [N] ha⁻¹ (N280) for the whole season. One half of the N fertilizer amount was applied as the basal-dressing at sowing and the other half was applied as the top-dressing at revival. Each treatment had three replicates with a randomized block design. The plant density was 500 plants m⁻². An irrigation of 50 mm was commonly carried out in March after winter wheat had turned green. In addition, other nutrient supply (e.g. phosphorus and potassium) and the cultivation practices, as well as the control of plant diseases, insect pests and weeds, were kept the same for all the plots.

2.2. Data collection

2.2.1. Biophysical measurements

Direct measurement of crop transpiration (T_c) is commonly difficult in the field. In this work, it was determined as the product of the reference evapotranspiration (ET_0) and the basal crop coefficient (K_{cb}) according to the FAO56 paper (Allen, Pereira, Raes, & Smith, 1998). The ET_0 was calculated using the FAO Penman-Monteith method from meteorological data. The K_{cb} of different stages were firstly looked up in Table 17 of Allen et al. (1998) and then adjusted according to the local conditions, cultural practices, soil water storage, crop leaf area index (LAI) and plant height information (for further details see Allen et al., 1998). Meteorological data for this study were recorded by a weather station installed at the experimental site. The soil moisture from the ground surface to a depth of 1.7 m was measured weekly by a neutron probe at a 0.1 m depth increments.

Plant height, green leaf area and aboveground biomass were measured every 10 days from revival to maturity. The plant height was determined by averaging 30 in-situ measurements for each plot. Crop samples were taken from 0.20 m of three random rows in each plot. The green leaf area of samples was measured with an LI-3000C Portable Area Meter (LI-COR, USA). Finally, the total sample (not including the roots) was oven-dried at 70 °C to a constant weight to obtain the aboveground biomass. Additional measurements were made at anthesis and maturity.

At maturity, three random samples of 1.0 m² were hand-harvested. Dry weights of the final aboveground biomass and the grain yield were recorded, and HI was obtained as the ratio of the grain yield to the total aboveground biomass.

2.2.2. Spectral measurements

The spectral reflectance of the winter wheat canopy was measured weekly from revival to maturity using the FieldSpec HandHeld Spectroradiometer (Analytical Spectral Devices, USA). This spectral device has a wavelength range of 325–1075 nm and a field of view of 25° full conical angle. The measurements were performed at 0.6 m above the crop canopy at noon on clear days. In each plot, 15 spectral samples were collected from three different locations at an optimized integration time and then averaged to reduce the possible field disturbances. Dark current and the reflectance of a white reference panel with known reflectance properties were

collected, following the operating procedures described by Analytical Spectral Devices (2002) strictly.

The vegetation index, NDVI, was calculated using the reflectance of the red band 620–670 nm and the near infrared band 841–876 nm, which is similar to the corresponding bands of the moderate-resolution imaging spectroradiometer (MODIS). The time-profile of NDVI from revival to maturity was obtained from the field measurements using cubic spline interpolation. The daily NDVI of the emergence–revival period was determined by linear interpolation between the first NDVI measurement at revival and the value of 0.1, which was the measured NDVI of the bare soil and assumed to be equal to the NDVI of the wheat crop at emergence.

2.2.3. Statistical analysis

Data were analyzed using SPSS (Statistical Product and Service Solutions) version 13.0. Correlation analysis was used to determine the relationships between HI and the biophysical and spectral variables. Binary regression was performed using linear and quadratic models to describe HI as a function of the cumulative NDVI after anthesis and the NDVI at anthesis. The regression coefficient was obtained according to the least squares method. For further evaluation of the relationships developed, the correlation coefficient (R), coefficient of determination (R^2) and the root mean square deviation (RMSD) were used. The RMSD was calculated using Eq. (7):

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (7)$$

where y_i and $\hat{y}_i$ was the measured and estimated HI of sample i , respectively, and N was the total sample number.

3. Results and discussion

3.1. Field measurements

Similar phenological development of winter wheat was observed in different N treatments in the same growing season, while there were some differences between the seasons due to weather factors. The key field parameters of 2008/2009 and 2009/2010 seasons are list in Table 1. In the 2008/2009 season, the total solar radiation was 2915 MJ m⁻² and the rainfall was 237.8 mm. There were 228 days between emergence and maturity, with 133 days before revival, 58 days between revival and anthesis and 37 days after anthesis, and the mean temperature of the three periods was 2.5 °C, 12.3 °C and 20.6 °C, respectively. In contrast, the 2009/2010 growth season had lower mean temperature of 1.1 °C before revival, which led to a delayed revival of winter wheat by 8 days and a shortening of revival–maturity period by almost a week. The rainfall in the 2009/2010 season totaled 110.8 mm with only 40.8 mm in the rapid growth period after revival, which was much less than for the previous season. The solar radiation of the 2009/2010 season was 2807 MJ m⁻². These differences between the two seasons would have effects on the crop biophysical variables, especially those sensitive to meteorological and water conditions such as crop transpiration (see below).

Table 1 – Main field parameters of the experimental site in the 2008/2009 and 2009/2010 wheat seasons.

	Main parameters	Emergence-revival	Revival-anthesis	Anthesis-maturity
2008/2009	Dates	25 Oct 2008–7 Mar 2009	7 Mar 2009–4 May 2009	4 May 2009–9 Jun 2009
	Number of days	133	58	37
	Mean temperature, °C	2.5	12.3	20.6
	Solar radiation, MJ m ⁻²	1158	1001	757
	Rainfall, mm	19.4	78.0	140.4
	Irrigation, mm	0.0	50.0	0.0
2009/2010	Dates	26 Oct 2009–16 Mar 2010	16 Mar 2010–10 May 2010	10 May 2010–12 Jun 2010
	Number of days	141	55	34
	Mean temperature, °C	1.1	11.8	20.7
	Solar radiation, MJ m ⁻²	1181	927	699
	Rainfall, mm	70.0	21.7	19.1
	Irrigation, mm	0.0	50.0	0.0

The main crop biophysical variables at different N levels, including the total transpiration (T_c), the transpiration after anthesis, the maximal LAI during the growing seasons (LAI_{max}), the aboveground biomass (AGB) at anthesis and maturity as well as the HI, are given in Table 2. Crop variables were greatly influenced by the N fertilization rates. In 2008/2009 season, crop transpiration for both the total season and after anthesis increased from low-N to high-N treatments with ranges of 198–328 mm and 71–122 mm respectively. Differences in crop transpiration among the N140, N210 and N280 treatments were not evident, which may be due to the similar LAI (Table 2) and the approximately full ground cover (generally defined as when $LAI \geq 3.0$, Allen et al., 1998) as well as the well-watered conditions. However, in the 2009/2010 season, the high transpiration rates of the N140, N210 and N280 treatments were limited by a lack of sufficient rainfall or water supply. The experimental results also showed that increasing the N fertilizer can increase the AGB and the HI, whereas excessive N fertilizer would not cause an increase but conceivably a stabilization or a decrease. Such an effect of super-optimal N application on the HI of wheat has been reported in the literature (e.g. Hay, 1995).

The NDVI time-profiles of different N treatments in the 2008/2009 season are shown in Fig. 1. During the vegetative growth stage of winter wheat from revival to heading, the canopy NDVI of all treatments increased rapidly until

reaching maximum values before heading, and then reduced significantly in the reproductive stage due to the decrease of canopy greenness. Among different N treatments, the canopy NDVI increased distinctly from low-N to high-N levels, but the changes became smaller with the increasing N fertilization rates (i.e. N140, N210 and N280), and there were similar variations in the above-mentioned crop biophysical parameters.

3.2. Estimating HI using the fraction of transpiration after anthesis

The fraction of water transpired after anthesis (θ_E or θ_{Ee}) was computed as the ratio of the cumulative transpiration in the anthesis-maturity period to that in the emergence-maturity period. Prior to the calculation of θ_{Ee} , the value of the transpiration should be normalized by the vapor pressure deficit according to Sadras and Connor (1991). The vapor pressure deficit was derived from the daily mean air temperature and the dew-point temperature. The measured values of HI against the θ_E and θ_{Ee} for the field experiments are presented in Fig. 2(a) and (b), respectively. Results showed that in the well-watered treatments, i.e. all the five N treatments in the 2008/2009 season and the N0 and N70 treatments in the 2009/2010 season, there were similar values of θ_E and θ_{Ee} . For the treatments of N140, N210 and N280 of the 2009/2010 season, which had some degree of water stress after anthesis,

Table 2 – Mean values of key biophysical variables of winter wheat in field experiments.

	Treatments	Total T_c , mm	T_c after anthesis, mm	LAI_{max} , m ² m ⁻²	AGB at anthesis, g m ⁻²	AGB at maturity, g m ⁻²	HI
2008/2009	N0	198	71	1.0	224	281	0.15
	N70	271	100	2.1	459	708	0.23
	N140	326	122	4.3	919	1131	0.39
	N210	323	120	4.6	1168	1705	0.43
	N280	328	122	4.9	1148	1434	0.44
2009/2010	N0	197	66	1.2	240	429	0.23
	N70	260	84	2.7	570	1036	0.35
	N140	235	57	3.6	757	1306	0.40
	N210	244	60	4.4	874	1428	0.43
	N280	266	72	4.7	872	1390	0.40

Notes: T_c , transpiration; LAI_{max} , maximal LAI; AGB, aboveground biomass; HI, harvest index; N0, N70, N140, N210, N280, different treatments respectively corresponding to N application rates of 0, 70, 140, 210, and 280 kg ha⁻¹.

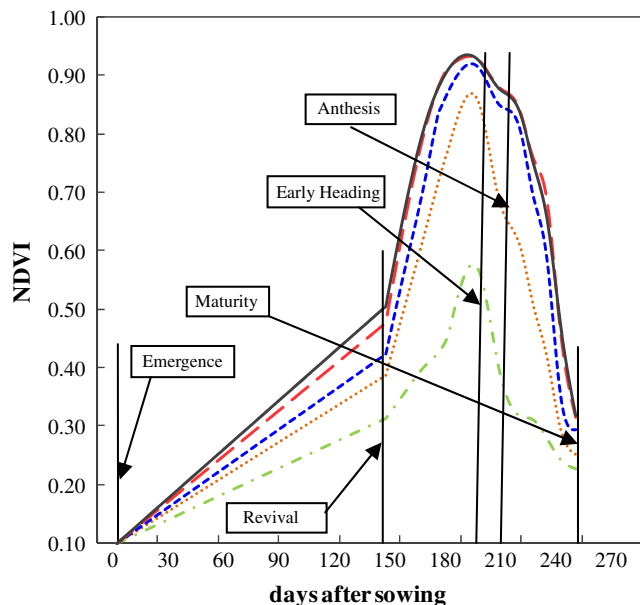


Fig. 1 – The daily NDVI time-profiles of winter wheat canopy at different N levels in 2008/2009 growth season. The treatments corresponding to the N application rates of 0, 70, 140, 210, and 280 kg ha⁻¹, are denoted as N0, N70, N140, N210 and N280, respectively. The initial dates of the main phenological stages are also given. — N0; N70; - - - N140; - · - N210; — N280.

the values of θ_E and θ_{Ee} decreased significantly. It was demonstrated that the θ_E and θ_{Ee} were not sensitive to the N deficiency but to the water shortage. As a result, no obvious relationship was observed between HI and θ_E or θ_{Ee} under the experimental conditions. For comparison, the Eq. (1) and Eq. (2), which were proposed by Richards and Townley-Smith (1987) and Sadras and Connor (1991) to describe the

relationships between HI and the fraction of transpiration after anthesis under drought conditions, are also shown in Fig. 2. Although estimating the HI as a function of θ_E or θ_{Ee} is effective for water-limited wheat (Sadras & Connor, 1991), it may be limited in well-watered conditions with other stresses, as is the case in this study.

3.3. Estimating HI using the fraction of biomass accumulation after anthesis

The measured HI and the ratio of the aboveground biomass produced after anthesis to that of the entire season (f_G) are presented in Fig. 3. There was almost no regular effect of N fertilizer on f_G . The observed data points were scattered and no evident relationship was found between HI and f_G under these experimental conditions. Such a result has been recorded previously in the literature. The relations of HI and f_G depicted by Eq. (3) and Eq. (4) in Kemanian et al. (2007), which are also presented in Fig. 3, were established based on the combined datasets of multi-year experiments of different treatments since the dataset within one year gave no clear changing trend of HI with f_G . In spite of the limited dataset in this study, the results suggest, at least, that estimating HI as a function of f_G had significant uncertainty and may result in large errors in practical application.

3.4. Estimating HI based on the canopy term of '1-NDVI_{post}/NDVI_{pre}'

The measured HI and the term '1-NDVI_{post}/NDVI_{pre}' of the 2008/2009 season, the 2009/2010 season and both seasons together are presented in Fig. 4(a), (b) and (c), respectively. The results showed that HI was significantly linearly correlated with the term '1-NDVI_{post}/NDVI_{pre}' in either of the two seasons, with a coefficient of determination of 0.82 and 0.72 respectively. This agreed with the description by Moriondo et al. (2007). However, there were obvious differences in the

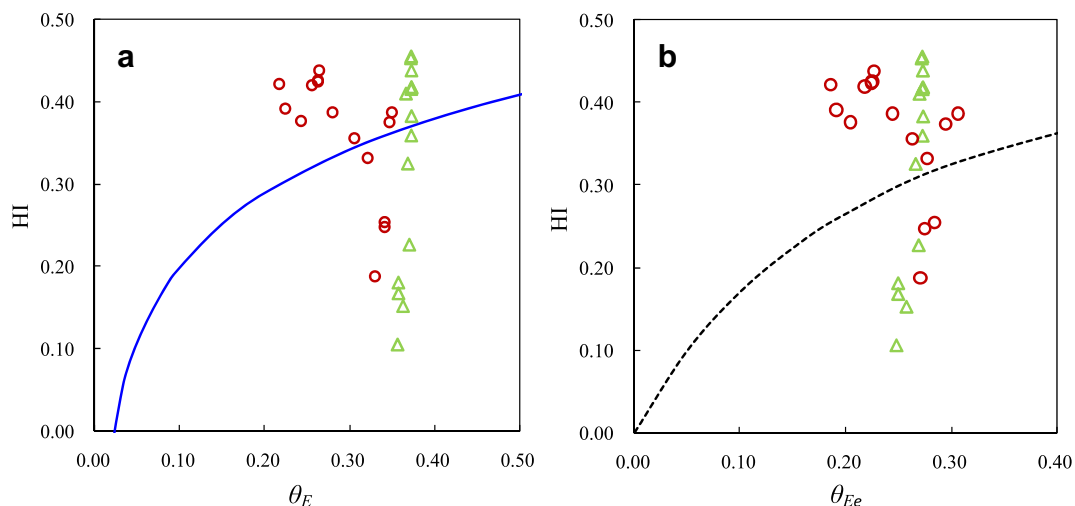


Fig. 2 – Relationship between the harvest index (HI) and the fraction of water transpired after anthesis (θ_E or θ_{Ee}) of winter wheat at different N levels. —, Eq. (1) with $a = -0.10$, $b = 0.13$ from Richards and Townley-Smith (1987); - - -, Eq. (2) with $a = 0.58$, $b = 1.68$ from Sadras and Connor (1991); Δ , measured values in 2008/2009 season; $\circ$, measured values in 2009/2010 season.

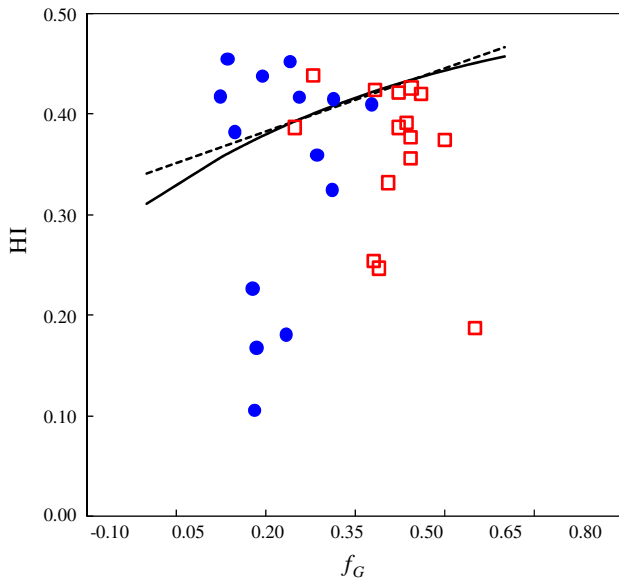


Fig. 3 – Relationship between harvest index (HI) and the fraction of biomass accumulation after anthesis (f_G) of winter wheat at different N levels. ●, observed data in 2008/2009 season; □, observed data in 2009/2010 season; —, Eq. (3) with $HI_0 = 0.34$, $s = 0.21$ by Kemanian et al. (2007); - - -, Eq. (4) with $HI_x = 0.52$, $HI_0 = 0.31$, $k = 1.99$ by Kemanian et al. (2007).

slopes and intercepts of the linear relationships between the current study and Moriondo et al. (2007), which may be caused by the differences in the datasets used in the two studies. The NDVI data in Moriondo et al. (2007) were derived from the NOAA-AVHRR images from satellite platform with the main phenological stages simulated by the crop model (CROPSYST), while in this work both the NDVI and the phenological data were obtained from in-situ field measurements. When using the measured data of the two seasons together, as shown in Fig. 4(c), a good correlation was also found between the HI and the ' $1 - NDVI_{post}/NDVI_{pre}$ ' with a coefficient of determination of 0.69. This suggested that HI

would not increase with the increasing ratio of $NDVI_{post}$ to $NDVI_{pre}$ without limit. When the $NDVI_{post}/NDVI_{pre}$ increased beyond a certain level, HI changed little and tended to a value close to its maximum. Thus it seems reasonable to use this approach to estimate the HI using the term ' $1 - NDVI_{post}/NDVI_{pre}$ ', for which the required data can be obtained by remote sensing over large areas.

3.5. Further analysis on the relationship between HI and the NDVI data of crop canopy

Following the concepts of fractional transpiration and fractional growth after anthesis (Kemanian et al., 2007; Sadras & Connor, 1991), the fraction of the cumulative NDVI after anthesis and its relationship to HI were analyzed. Similar to f_G and θ_E or θ_{Ee} , the fraction of cumulative NDVI after anthesis was defined as the ratio of the cumulative NDVI of the anthesis-maturity period to that of the emergence-maturity period. The measured HI and the fraction of the cumulative NDVI after anthesis are presented in Fig. 5(a). Lower HI corresponded to a lower fraction of the cumulative NDVI after anthesis, and the relationship can be described by a natural logarithm with a correlation coefficient of 0.79. It was demonstrated that using the NDVI data to estimate HI based on the partitioning of crop activity before and after flowering was feasible since strong correlations were found between them. However, due to the influences of weather conditions, some difficulties may be encountered in acquiring long-term NDVI data of winter wheat from emergence to maturity in practical applications.

In view of the fact that the period from anthesis to maturity is closely related to grain formation, the possibility of estimating HI using the short-term NDVI dataset of this period was further investigated in this study. The measurements of the crop HI and the cumulative NDVI of the anthesis-maturity period are presented in Fig. 5(b). It was found that HI was significantly related to the cumulative NDVI after anthesis, with a high correlation coefficient of 0.87. It seemed workable to estimate HI just using the canopy NDVI data after anthesis. If so, it would be more favorable to practical application because it requires a shorter dataset.

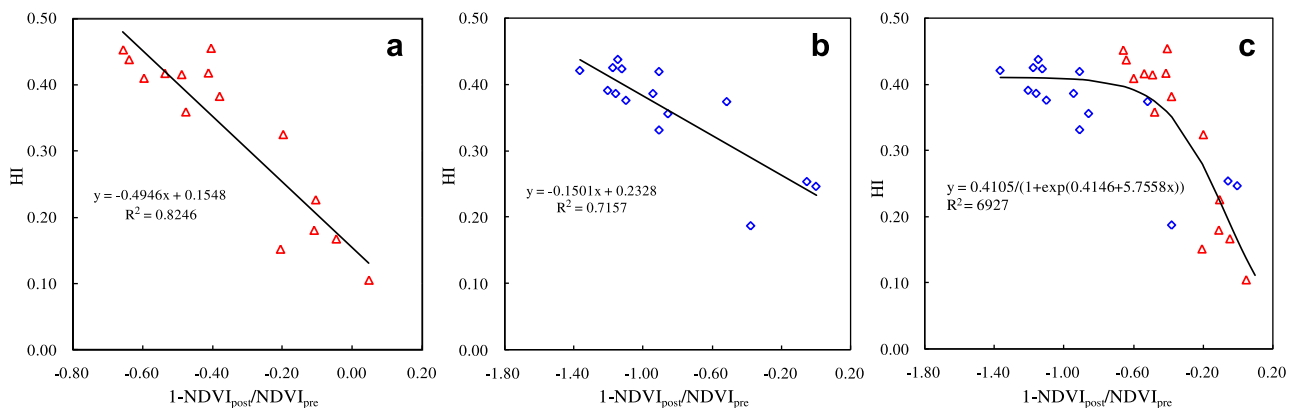


Fig. 4 – Relationship between the harvest index (HI) and the term ' $1 - NDVI_{post}/NDVI_{pre}$ ' of winter wheat at different N levels. The $NDVI_{pre}$ and $NDVI_{post}$ denote the mean NDVI values of the pre- and post-anthesis period respectively. Δ, measured values in 2008/2009 season; ◇, measured values in 2009/2010 season; —, regression line of the measured dataset.

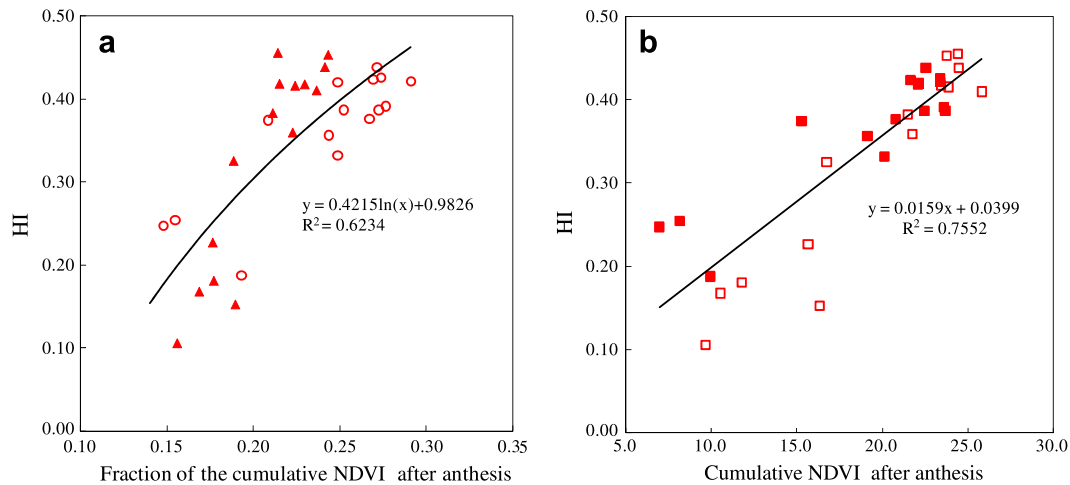


Fig. 5 – Dependency of harvest index (HI) on the fraction of cumulative NDVI after anthesis (a), and on cumulative NDVI after anthesis (b). ▲, measured data from 2008/2009 season in subfigure (a); ○, measured data from 2009/2010 season in subfigure (a); □, measured data from 2008/2009 season in subfigure (b); ■, measured data from 2009/2010 season in subfigure (b); —, regression line of the measured dataset.

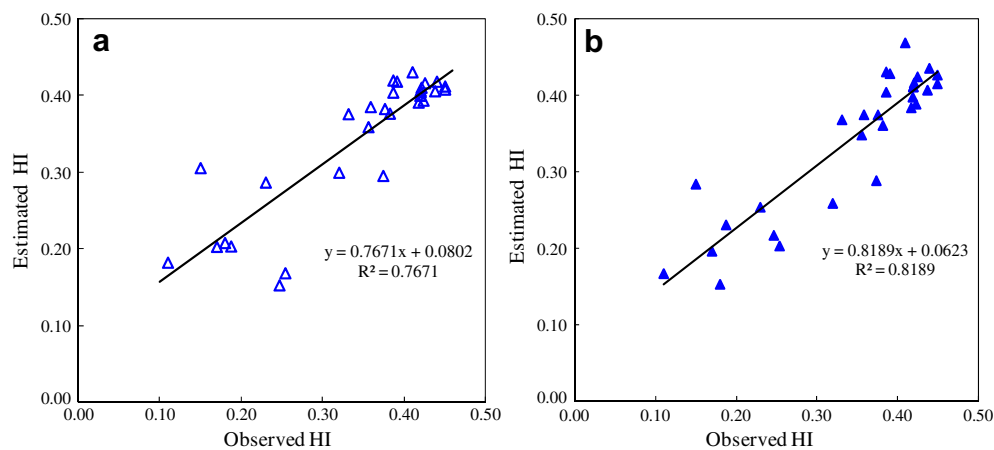


Fig. 6 – Estimated harvest index (HI) by binary regression analysis vs the measured HI of the 2008/2009 and 2009/2010 wheat seasons. Subfigure (a) presents binary linear regression and (b) shows binary quadratic regression based on the NDVI at anthesis and the cumulative NDVI after anthesis. The thick line is regression line and the light line is the 1:1 relationship line. The root mean square deviation (RMSD) is 0.05 for (a) and 0.04 for (b). The coefficient of determination between the estimated and observed HI is 0.77 and 0.82 for (a) and (b) respectively.

The above relationship between HI and cumulative NDVI after anthesis reflected the dominant effect of the crop post-anthesis growth on the HI. Considering that the crop growth status before flowering can also influence the HI, the NDVI at anthesis was introduced to represent the crop growth from emergence to anthesis. Binary regression analysis was performed with the linear and quadratic forms based on the measurements of HI, NDVI at anthesis and the cumulative NDVI after anthesis. The correlation coefficient of the binary linear and quadratic regression relationship was respectively 0.88 and 0.90, which was an improvement on the single-element regressions. The estimated values of HI by binary linear and quadratic regressions, as well as the measured HI, are given in Fig. 6(a) and (b), respectively. The 1:1 relationship

line is also shown for comparison. The RMSD of the estimated HI was 0.05 for binary linear regression and 0.04 for binary quadratic regression, with $R^2 = 0.77$ and 0.82, respectively. The results demonstrated that the short-term NDVI dataset from anthesis to maturity can give a good estimation of the crop HI, especially using the binary regression relations of HI, NDVI at anthesis and cumulative NDVI after anthesis.

4. Conclusions

Nitrogen (N) fertilizer had great effects on HI, canopy NDVI, crop transpiration and the aboveground biomass of winter wheat. The fraction of water transpired after anthesis (θ_E or θ_{Ee}) was not

sensitive to the N fertilization rates under well-watered conditions, and thus using this parameter to estimate the HI may be limited in growing conditions with adequate water supply, as was the case in this study. The relationship between HI and the fraction of the aboveground biomass accumulation after anthesis (f_G) had significant uncertainty and thus it may result in large errors in HI estimation. Strong correlations were found between HI, the term ' $1 - \text{NDVI}_{\text{post}}/\text{NDVI}_{\text{pre}}$ ' and the fraction of the cumulative NDVI after anthesis, so it seemed workable to estimate HI using the canopy NDVI data based on the crop partitioning activity before and after flowering. But it may be difficult to obtain the long-term NDVI data of winter wheat from emergence to maturity in practice due to the changing weather conditions.

Further analysis demonstrated that HI was highly related to the cumulative NDVI after anthesis, and that a high accuracy of HI estimation can be achieved just using the short-term dataset of canopy NDVI from anthesis to maturity, especially by introducing another variable of NDVI at anthesis via binary regression analysis.

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